

① $X \sim N(\mu, \sigma^2)$... under $\mathbb{P}$

$\theta X \begin{cases} \text{mean} = \theta \mu \\ \text{variance} = \theta^2 \sigma^2 \end{cases}$

$$M_X(\theta) = \mathbb{E}[e^{\theta X}] = e^{\text{mean} + \frac{1}{2} \text{variance}} \\ = e^{(\theta \mu + \frac{1}{2} \theta^2 \sigma^2)}$$

$$\frac{dM}{d\theta} = e^{(\theta \mu + \frac{1}{2} \theta^2 \sigma^2)} \times (\mu + \theta \sigma^2)$$

$$\theta=0, \quad \frac{dM}{d\theta} = \mu$$

$$\frac{d^2 M}{d\theta^2} = e^{(\theta \mu + \frac{1}{2} \theta^2 \sigma^2)} \times (\sigma^2) + (\mu + \theta \sigma^2)^2 \times e^{(\theta \mu + \frac{1}{2} \theta^2 \sigma^2)}$$

$$\theta=0, \quad \frac{d^2 M}{d\theta^2} = \sigma^2 + \mu^2 = \mathbb{E}(X^2)$$

$$\sigma^2 = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \sigma^2 + \mu^2 - \mu^2 = \sigma^2$$

$$X_1 \sim N(\mu_1, \sigma_1^2) \dots \text{under } \mathbb{P}$$

$$X_2 \sim N(\mu_2, \sigma_2^2) \dots \text{under } \mathbb{P}$$

$$Y = X_1 + X_2$$

$$M_Y(\theta) = \mathbb{E}(e^{\theta Y})$$

$$= \mathbb{E}(e^{\theta(X_1 + X_2)})$$

$$= \mathbb{E}(e^{\theta X_1}) * \mathbb{E}(e^{\theta X_2}) \dots X_1, X_2 \text{ are independent}$$

$$= \left(e^{\theta \mu_1 + \frac{1}{2} \theta^2 \sigma_1^2} \right) \left(e^{\theta \mu_2 + \frac{1}{2} \theta^2 \sigma_2^2} \right)$$

$$= e^{\theta(\underline{\mu_1 + \mu_2}) + \frac{1}{2} \theta^2 (\underline{\sigma_1^2 + \sigma_2^2})}$$

$$Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

$$\frac{dQ}{dP} = e^{-\lambda W_T - \frac{1}{2} \lambda^2 T}$$

$W_T \xrightarrow{\quad} \underline{N(0, T)} \text{ under } \mathbb{P}$
 $\searrow \quad \quad \quad ?? \text{ under } \mathbb{Q}$

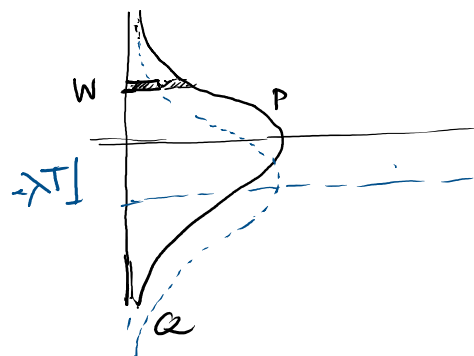
$$\begin{aligned} \mathbb{E}^{\mathbb{Q}}(e^{\theta W_T}) &= \mathbb{E}^{\mathbb{P}}(e^{\theta W_T} \cdot e^{-\lambda W_T - \frac{1}{2} \lambda^2 T}) \\ &= \mathbb{E}^{\mathbb{P}}\left(e^{\frac{(\theta - \lambda) W_T - \frac{1}{2} \lambda^2 T}{x}}\right) = e^{-\frac{1}{2} \lambda^2 T + \frac{1}{2} (\theta - \lambda)^2 T} \\ &= e^{-\theta \lambda T + \frac{1}{2} \theta^2 T} \end{aligned}$$

$$\begin{aligned} X &= (\theta - \lambda) W_T - \frac{1}{2} \lambda^2 T \\ \mathbb{E}(X) &= \underline{\underline{-\frac{1}{2} \lambda^2 T}}, \quad \mathbb{V}(X) = \underline{\underline{(\theta - \lambda)^2 T}} \end{aligned}$$

$\sim N(-\lambda T, T)$

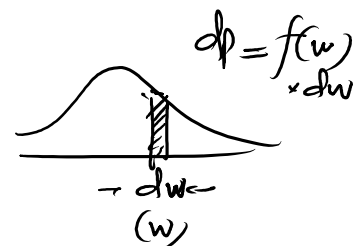
$$\underline{W_T^{\mathbb{Q}}} \sim N(-\lambda T, T)$$

$$\underline{W_t^{\mathbb{Q}}} \sim N(-\lambda t, t)$$



$$w_T^P \sim \underline{N(0, T)}$$

$$w_T^Q \sim \underline{N(-\lambda T, T)}$$



$$\begin{aligned} \frac{dq}{dp} &= \left(\frac{dq}{dw} \right) \left(\frac{dw}{dp} \right) = \left(\frac{dq}{dw} \right) / \left(\frac{dp}{dw} \right) \rightarrow \\ &= \frac{f^Q(w)}{f^P(w)} = \frac{\frac{1}{\sqrt{2\pi T}} e^{-(w_T + \lambda T)^2 / 2T}}{\frac{1}{\sqrt{2\pi T}} e^{-w_T^2 / 2T}} \\ &= e^{\frac{w_T^2}{2T} - (w_T + \lambda T)^2 / 2T} = e^{-\lambda w_T - \frac{1}{2} \lambda^2 T} \end{aligned}$$

$$\textcircled{1} \quad X_t = \mu t + \sigma W_t, \quad W_t^{\mathbb{P}} \sim N(0, t)$$

$$\text{Target: } X_t = \sigma \tilde{W}_t, \quad \tilde{W}_t^{\mathbb{Q}} \sim N(0, t)$$

$$\sigma \tilde{W}_t = \mu t + \sigma W_t$$

$$\Rightarrow \tilde{W}_t = \underline{W_t} + \underline{\left(\frac{\mu}{\sigma}\right)t} \quad , \quad \lambda = \mu/\sigma$$

$$\frac{dQ}{dP} = e^{-(\mu/\sigma)t - \frac{1}{2}(\mu^2/\sigma^2)t}$$

$$\textcircled{2} \quad \underline{dS_t} = \mu S_t dt + \sigma S_t dW_t, \quad W_t^{\mathbb{P}} \sim N(0, t)$$

$$\underline{dS_t} = r S_t dt + \sigma S_t d\tilde{W}_t, \quad \tilde{W}_t^{\mathbb{Q}} \sim N(0, t)$$

$$\cancel{\mu S_t dt} + \cancel{\sigma S_t dW_t} = \cancel{r S_t dt} + \cancel{\sigma S_t d\tilde{W}_t}$$

$$\Rightarrow d\tilde{W}_t = \left(\frac{\mu - r}{\sigma} \right) dt + dW_t$$

$$\tilde{W}_t = \underbrace{\left(\frac{\mu - r}{\sigma} \right) t}_{\lambda} + W_t$$

$$\frac{dQ}{dP} = e^{\left(-\lambda W_t - \frac{1}{2} \lambda^2 t \right)}$$

$$\frac{dq}{dp} = e^{(w_t - \frac{1}{2}t)}$$

$$e^{(-\lambda w_t - \frac{1}{2}\lambda^2 t)}$$

$$\lambda = (-1)$$

$$\tilde{w}_t = w_t - t$$

$$Z_t = \frac{dQ}{dP} = e^{(-\lambda W_t - \frac{1}{2}\lambda^2 t)}$$

$$Z_t = \mathbb{E}^P(\underline{Z}_t^f | \mathcal{F}_t), \quad \mathbb{E}^P(\mathbb{E}^P(\underline{Z}_t | \mathcal{F}_t) | \mathcal{F}_s)$$

$$Z_0 = ?? \quad (1) \quad \mathbb{E}^P(\underline{Z}_t | \mathcal{F}_0) = 1 \quad \xleftarrow{\text{Tower}}$$

$$\mathbb{E}^P(\underline{Z}_t | \mathcal{F}_s) = \underline{Z}_s$$

$$\begin{aligned} &= \mathbb{E}^P(e^{-\lambda W_t - \frac{1}{2}\lambda^2 t} | \mathcal{F}_s) = \mathbb{E}^P(e^{-\lambda(W_t - W_s + W_s) - \frac{1}{2}\lambda^2(t-s+s)} | \mathcal{F}_s) \\ &\quad \mathcal{F}_0 = \mathbb{E}^P(e^{-\lambda W_s - \frac{1}{2}\lambda^2 s}) \times \mathbb{E}^P(e^{-\lambda(W_t - W_s) - \frac{1}{2}\lambda^2(t-s)} | \mathcal{F}_s) \\ &= \underline{Z}_s \times e^{-\cancel{\frac{1}{2}\lambda^2(t-s)} + \cancel{\frac{1}{2}\lambda^2(t-s)}} = \underline{Z}_s \end{aligned}$$

$$\underline{\underline{\zeta_t = e^{-\lambda W_t - \frac{1}{2}\lambda^2 t}}}$$

$$\underline{\underline{\frac{\partial \zeta_t}{\partial t} = \zeta_t \times \left(-\frac{1}{2}\lambda^2\right)}}$$

$$\underline{\underline{\frac{\partial \zeta_t}{\partial W_t} = \zeta_t \times (-\lambda)}}$$

$$\underline{\underline{\frac{\partial^2 \zeta_t}{\partial W_t^2} = \zeta_t \cdot (\lambda^2)}}$$

$$d\zeta_t = \cancel{\zeta_t \cdot \left(-\frac{1}{2}\lambda^2\right) dt} + \zeta_t (-\lambda) dW_t + \cancel{\frac{1}{2} \zeta_t (\lambda^2) \cdot dt}$$

$$\underline{\underline{\langle d\zeta_t = -\lambda \zeta_t dW_t \rangle}}$$